DSBDL Assignment 02 - Data Wrangling 2

Create an "Academic performance" dataset of students and perform the following operations using Python.

1. **Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.**
2. **Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.**
3. **Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution. Reason and document your approach properly.**

Dataset details: https://www.kaggle.com/datasets/spscientist/students-performance-in-exams

In [1]:

```
import numpy as np
import pandas as pd
import seaborn as sns

sns.set_theme( rc={ "figure.figsize": (5,3) } )
```

In [2]:

```
ds = pd.read_csv( "dataset.csv" )
ds.head()
```

Out[2]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

In [3]:

```
ds = ds.rename( columns={
    "race/ethnicity": "race" ,
    "parental level of education": "parent_edu" ,
    "test preparation course": "course_completed" ,
    "math score": "score_math" ,
    "reading score": "score_reading" ,
    "writing score": "score_writing"
} )
ds.head()
```

Out[3]:

	gender	race	parent_edu	lunch	course_completed	score_math	score_reading	score_writing
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44

	gender	race	parent_edu	lunch	course_completed	score_math	score_reading	score_writing
4	male	group C	some college	standard	none	76	78	75

In [4]:

```
ds.dtypes
```

Out[4]:

```
gender               object
race                 object
parent_edu           object
lunch                object
course_completed     object
score_math            int64
score_reading         int64
score_writing         int64
dtype: object
```

1. Checking for missing values

In [5]:

```
ds.isna().sum()
```

Out[5]:

```
gender               0
race                 0
parent_edu           0
lunch                0
course_completed     0
score_math           0
score_reading        0
score_writing        0
dtype: int64
```

2. Analysis of numeric variables: `score_math`, `score_reading` and `score_writing`

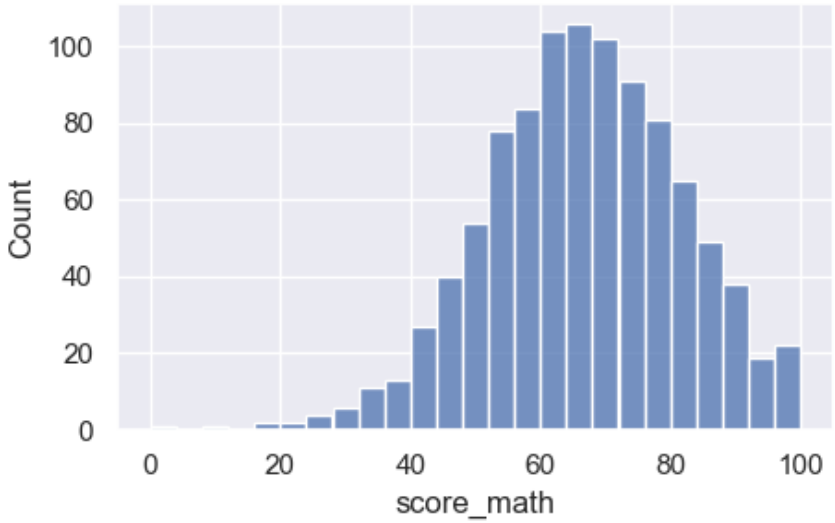
2.1. Visualizing distributions of numeric variables

In [6]:

```
sns.histplot( ds.score_math )
```

Out[6]:

```
<Axes: xlabel='score_math', ylabel='Count'>
```

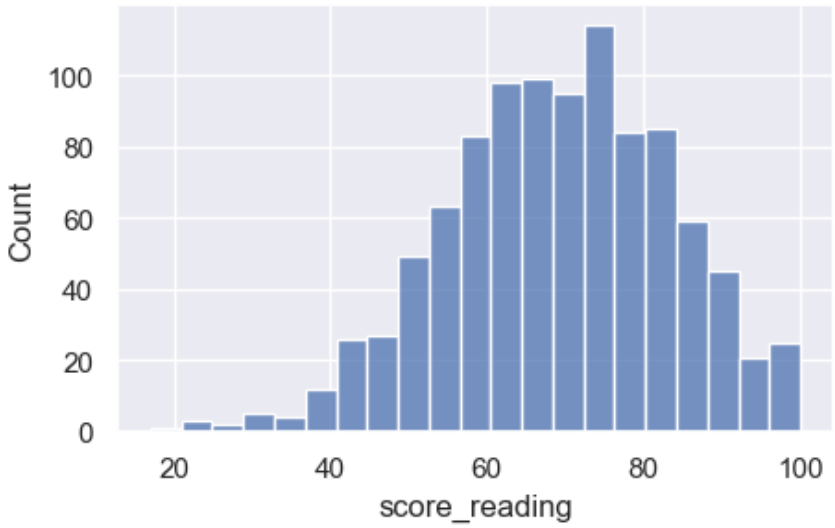


In [7]:

```
sns.histplot( ds.score_reading )
```

Out[7]:

```
<Axes: xlabel='score_reading', ylabel='Count'>
```

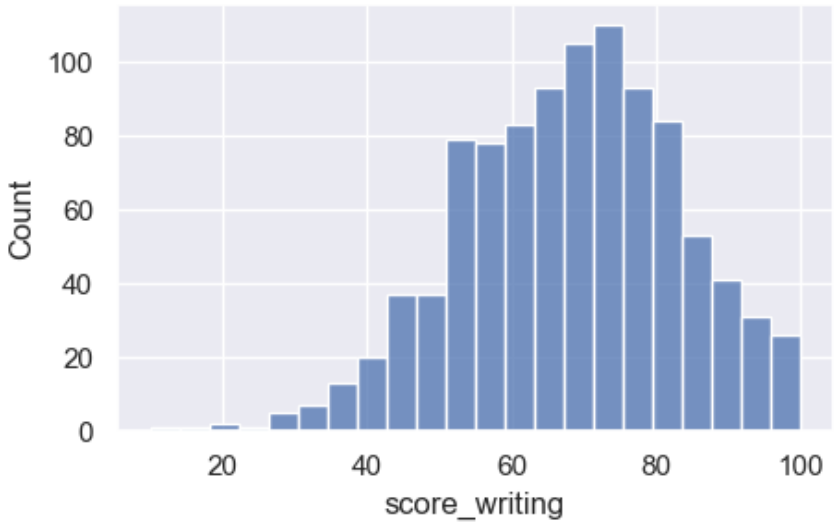


In [8]:

```
sns.histplot( ds.score_writing )
```

Out[8]:

```
<Axes: xlabel='score_writing', ylabel='Count'>
```



2.2. Remove outliers with IQR Rule

In [9]:

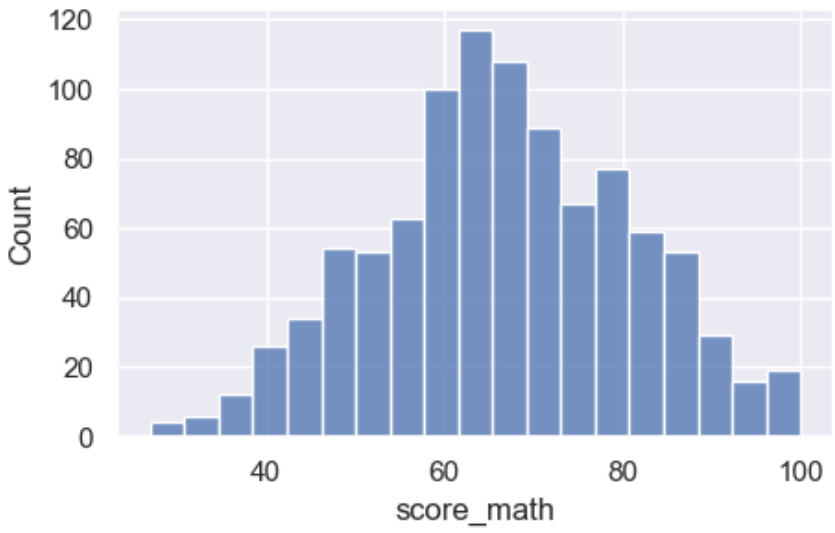
```
def remove_outliers(feature):
    global ds
    q3 , q1 = np.percentile( ds[feature] , [ 75 , 25 ] )
    iqr = q3 - q1
    ds = ds[ (ds[feature] >= q1 - 1.5 * iqr) & (ds[feature] <= q3 + 1.5 * iqr) ]

remove_outliers( "score_writing" )
remove_outliers( "score_math" )
remove_outliers( "score_reading" )
```

In [10]:

```
sns.histplot( ds.score_math )
```

Out[10]:

```
<Axes: xlabel='score_math', ylabel='Count'>
```

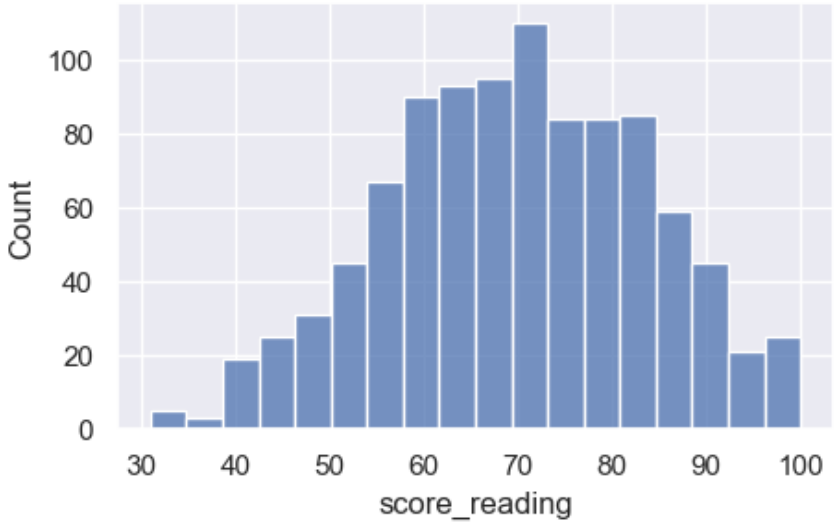


In [11]:

```
sns.histplot( ds.score_reading )
```

Out[11]:

```
<Axes: xlabel='score_reading', ylabel='Count'>
```

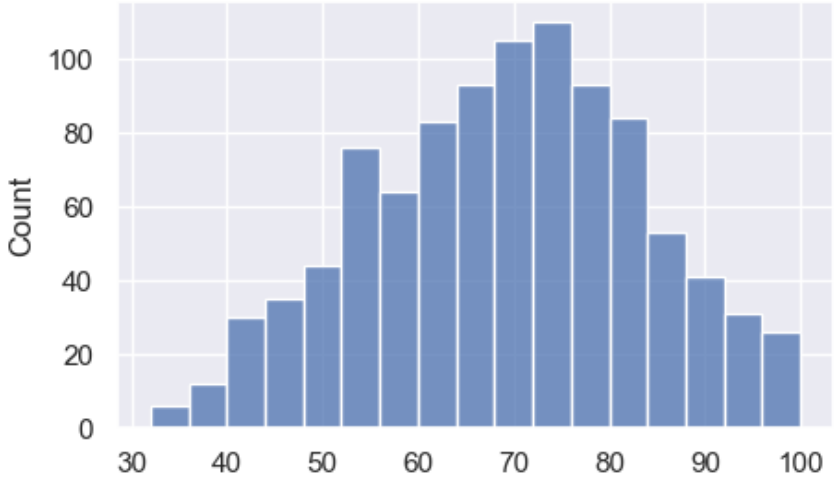


In [12]:

```
sns.histplot( ds.score_writing )
```

Out[12]:

```
<Axes: xlabel='score_writing', ylabel='Count'>
```



score_writing

3. Data Transformation

3.1. Encoding categorical variables

In [13]:

```
def encode_categorical(feature):
    values = ds[feature].unique()
    label = 0
    for val in values:
        ds.loc[ ds[feature] == val , feature ] = label
        label += 1

encode_categorical( "gender" )
encode_categorical( "race" )
encode_categorical( "parent_edu" )
encode_categorical( "lunch" )
encode_categorical( "course_completed" )
```

3.2. Normalizing numeric features

In [14]:

```
# As outliers are eliminated, both min-max or z-score
# normalization can be used
# Ref: https://stats.stackexchange.com/questions/547446/z-score-vs-min-max-normalization
def z_score( feature ):
    ds[feature] = ( ds[feature] - ds[feature].mean() ) / ds[feature].std()

z_score( "score_reading" )
z_score( "score_writing" )
z_score( "score_math" )
```

In [15]:

```
ds
```

Out[15]:

	gender	race	parent_edu	lunch	course_completed	score_math	score_reading	score_writing
0	0	0	0	0	0	0.369943	0.163678	0.370964
1	0	1	1	0	1	0.160750	1.457644	1.341360
2	0	0	2	0	0	1.625105	1.817079	1.687930
3	1	2	3	1	0	-1.373337	-0.914627	-1.708457
4	1	1	1	0	0	0.648868	0.595000	0.440278
...	...	...	...	...	...	...	...	...
995	0	4	2	0	1	1.485643	2.104627	1.826558
996	1	1	4	1	0	-0.327369	-1.058402	-0.946003
997	0	1	4	1	1	-0.536563	0.091791	-0.252863
998	0	3	1	0	1	0.091018	0.595000	0.578906
999	0	3	1	1	0	0.718599	1.170096	1.202732

986 rows × 8 columns